BigPrint TV — how to try it

Five minutes with a Vega virtual device; fifteen seconds without one. No AWS account, no sign-in. Backend live to 20 November 2026.

What it is. Photograph a letter or bill with any phone. It appears on the television in very large print, with the figures that matter pinned at the top, and one press reads it aloud. Built for people with age-related sight loss, on the screen they already own.

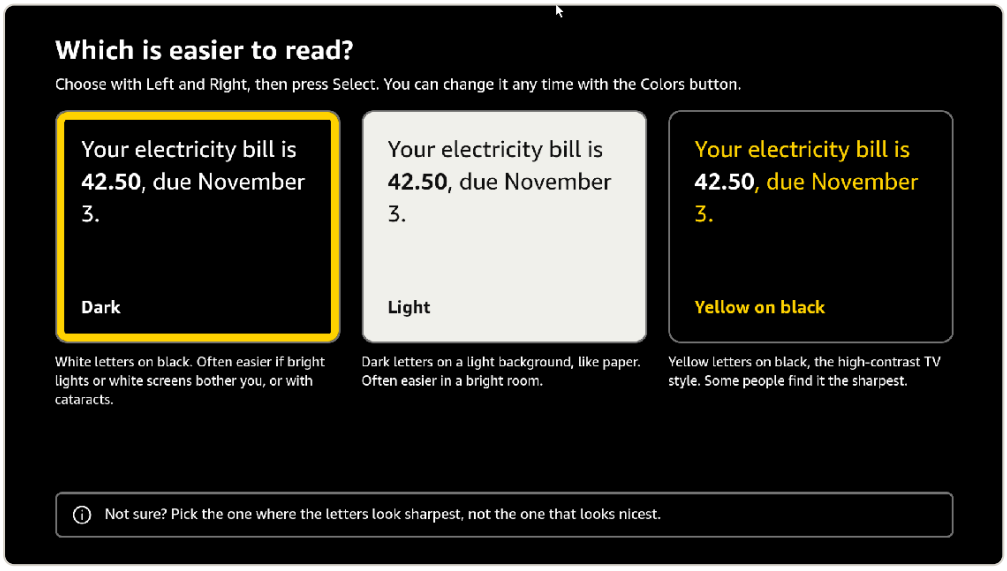
1. Install

You need the Vega SDK with its virtual device, or a Fire TV running Vega OS in developer mode. Download the package for your target from `github.com/rukhsarai/bigprint-tv/releases/latest` (also in the Devpost upload): `x86_64` for the virtual device, `aarch64` for a Fire TV.

```
vega virtual-device start                      # skip on a real Fire TV
vega install-app bigprint-tv-0.1.0_x86_64.vpkg
vega run-app com.bigprint.tv.main
```

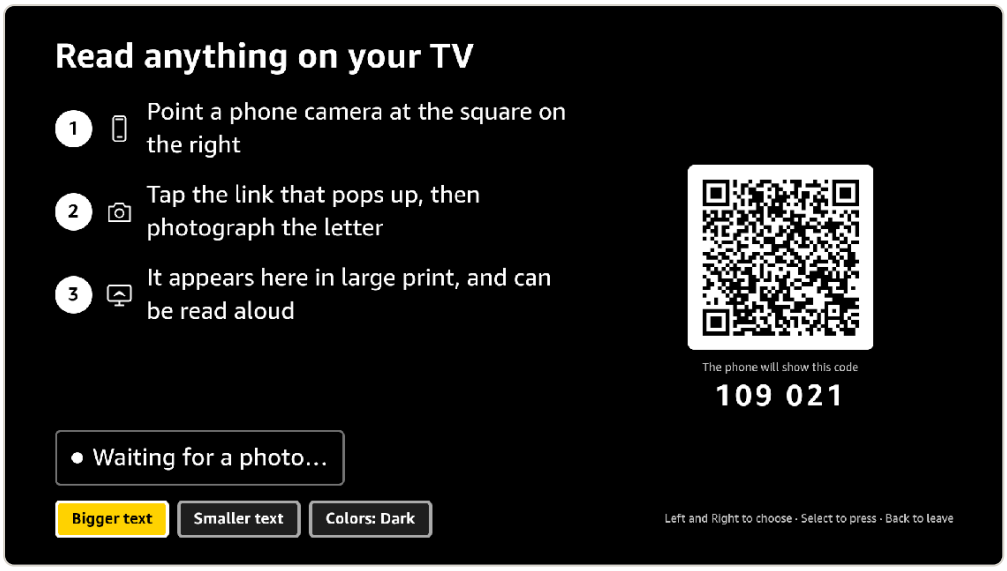
2. Run it, step by step

1. **Pick a look.** The app opens on the same sentence in dark, light, and yellow on black. Move with the D-pad, press Select on the one you can read best.



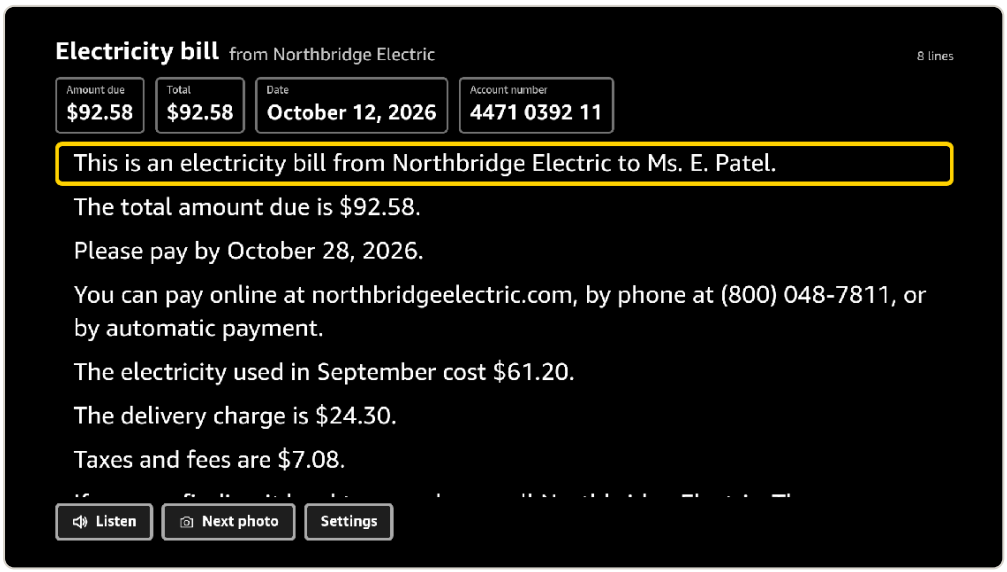
Step 1 — pick a look by reading a real sentence, not by name.

2. **Scan the QR code.** The TV shows a QR code and a six-digit code. Scan it with any phone's camera; a web page with one button opens. Nothing to install on the phone.



Step 2 — the pairing screen. Any phone's camera opens it.

3. **Photograph the bill.** Open `samples/us-bill.jpg` on a laptop screen and photograph that, or print it. Hold the phone square to the page, fill the frame, press the one button.
4. **Read it on the TV.** The TV says *Reading your letter*, and about ten seconds later the page is on screen. Move line by line with the D-pad; the focused line has a thick yellow ring.

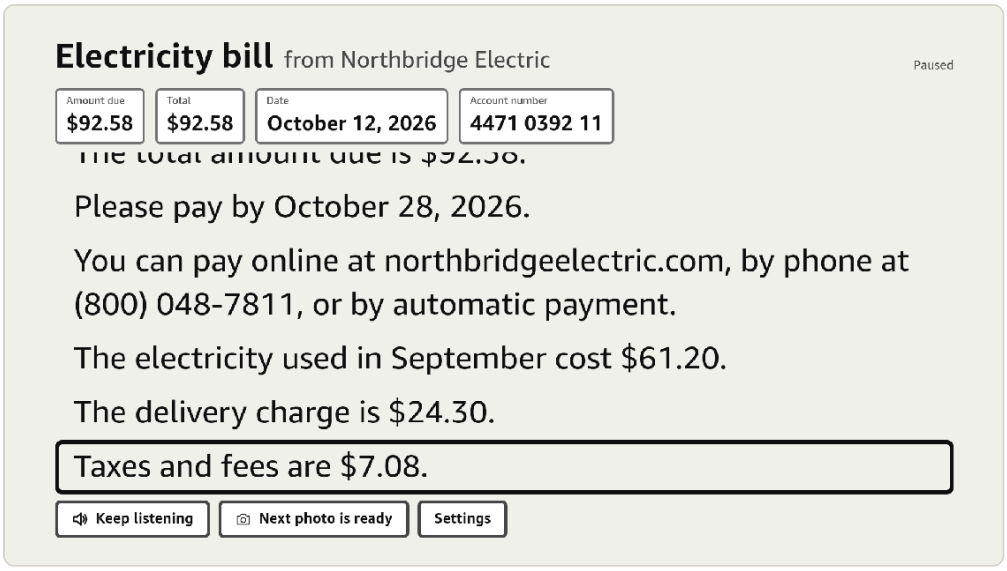


Step 4 — the key figures pinned above the letter; the ringed line is the focus.

> **Check the TV against the paper.** It should show amount due **$92.58**, due date **October 28, 2026**, statement date October 12, 2026, account **4471 0392 11**. Every one of those came from OCR. A figure on the TV that is not on the paper is a bug worth reporting.

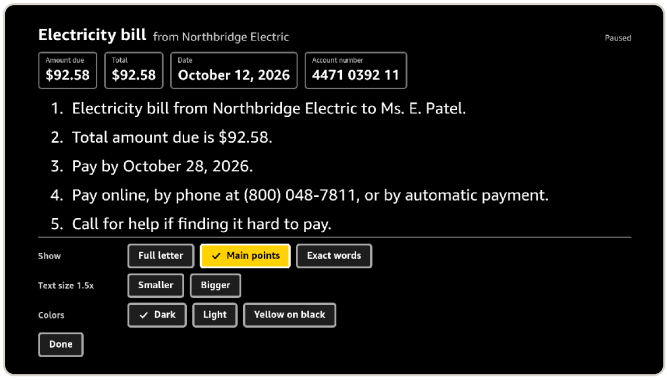
3. Four things to try

1. **Listen.** Press the first button at the bottom. The whole page is read aloud in a natural voice and the button becomes *Pause*. The voice is made only on request, which is why the text arrives first.

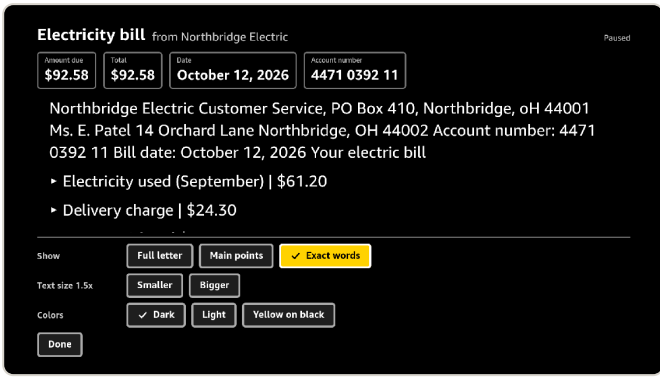


Listening, paused. While you read, the next photo can already be waiting — the middle button says so.

2. **Three ways to read it.** *Settings* → *Show* switches between the whole letter in plain English, just the main points, and the exact words as printed, with table rows kept as rows.

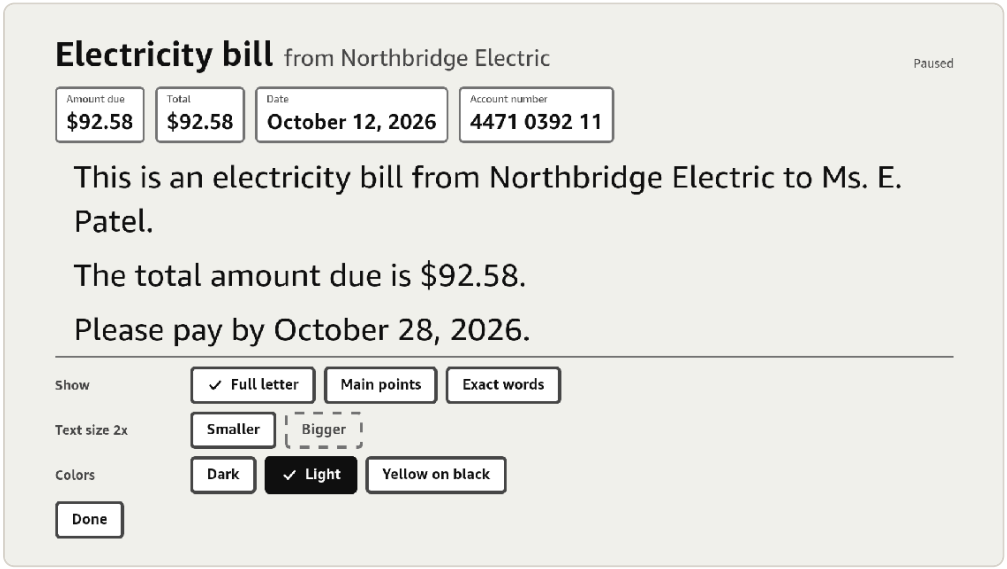


Main points.



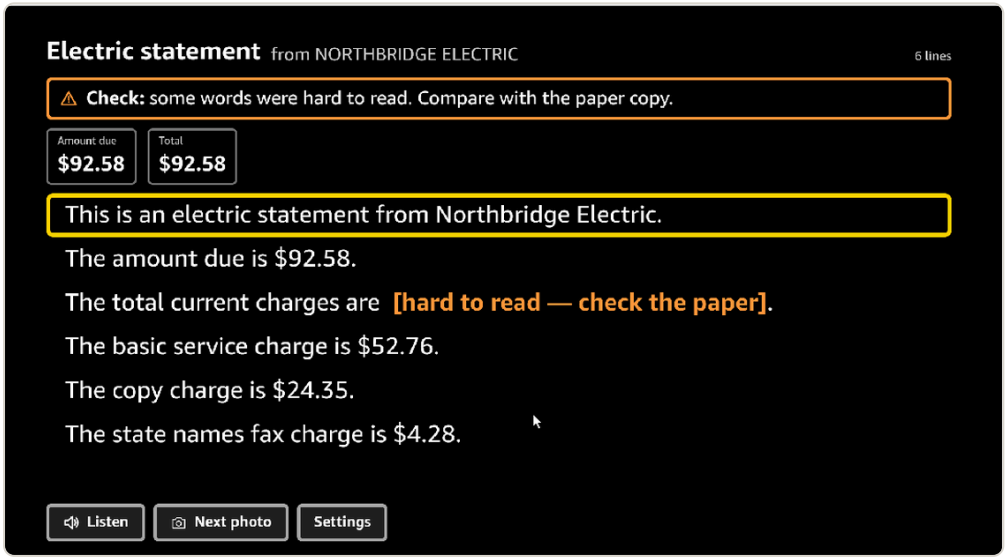
Exact words.

3. **Text size and looks.** *Settings* → *Text size* goes up to 2× and the pinned figures never scroll away; *Settings* → *Look* switches dark, light, and yellow on black. Every colour pair passes WCAG 7:1, checked in the test suite.



The light look at 2× text: the pinned figures stay put while the letter grows.

4. **A page that is hard to read.** Photograph `samples/us-bill-very-blurry.jpg`, the same bill blurred until the OCR starts to struggle. Three things happen, and the third is the one to look at: a *Check* line appears above the letter; only the figures the pipeline is sure of are pinned; and where a money figure fell below the confidence floor, the sentence keeps its shape and the figure is replaced by **[hard to read — check the paper]**. The reading is not shown at all, because a figure that was not good enough to pin is not good enough to say.



The badly blurred page. The amount due is pinned because both readers agree on it; the total current charges are not stated at all, because that reading was not trustworthy. Press Listen and the voice says the same thing in words.

Next photo and Recent letters. *Next photo* returns to the pairing screen; the phone page is still open, so it is one more tap. Press *Back* on the remote from the letter for *Recent letters*, held on the TV for a day and then deleted. Back never exits the app by accident.

4. Without a device

From a terminal with `curl` and Python, in the repository:

```
backend/tools/round_trip.sh                    # live backend, end to end, ~15 s
backend/tools/round_trip.sh samples/us-bill.jpg # the same with the video's bill

npm ci && npm test                             # app: 162 tests
python3 -m unittest discover backend/tests     # backend: 44 tests
```

It opens a pairing session, uploads a photo, waits for the text, asks for the voice, and prints the key facts the pipeline extracted with the time each step took. That is the whole product minus the screen.

5. What is going on underneath

STEP	SERVICE	RULE
Read the photo, with layout and tables	Amazon Textract	A bill's rows stay rows
Classify the page	Amazon Nova Lite	Bill, letter, receipt, certificate
Pull the money figures	Textract AnalyzeExpense	Numbers come from OCR, never from a model
Second look at the photo	Amazon Nova Pro	May veto a figure; can never add one
Rewrite into spoken sentences	Amazon Nova Pro	Any sentence with a number not on the page is dropped
Voice, on request	Amazon Polly	Made only when someone presses Listen

Before you photograph a real letter. Photos go to AWS (us-east-1) to be read; photo, text and audio are deleted after one day. It is a hackathon deployment, so use the samples or a letter you do not mind a demo server seeing. There is a cap of 200 pages a day across everyone. **Known limits:** no run yet on a physical Fire TV with VoiceView, and a pairing-screen overflow at the largest text sizes.